Proof by induction- series

Basic skills

Q1. seq $a_1=1, a_2=2, a_3=2, a_4=-1$

a) Partial sum $S_1=a_1=1$

b) $S_2=a_1+a_2=1+2=3$

c) $S_3=a_1+a_2+a_3=1+2+2=5$

d) $S_4=\sum_{r=1}^{4} a_r = 1+2+2+1=6$

Q2. $a_n=n$

a) $S_1=a_1=1$, b) $S_2=a_1+a_2=1+2=3$

c) $S_3=1+2+3=6$ d) $S_5=\sum_{r=1}^{5} r = \frac{5\times6}{2}=15$

e) $S_{10}=\sum_{r=1}^{10} r = \frac{10\times11}{2}=55$

Q3. a) $S_1=\frac{1}{3}$, $f(1)=\frac{1}{2\times1+1}=\frac{1}{3}$ ✓

$S_2=\frac{1}{3}+\frac{1}{15}=\frac{6}{15}$ $f(2)=\frac{2}{2\times2+1}=\frac{2}{5}$ ✓

$=\frac{2}{5}$

$S_3 = \frac{1}{3} + \frac{1}{15} + \frac{1}{35} = \frac{3}{7}$ $\quad f(3) = \frac{3}{2\times3+1} = \frac{3}{7}$ ✓

b) $S_1 = f(1)$ is the base case

c) we would have to show

$$S_{k+1} = f(k+1)$$

given $S_k = f(k)$ ← I.H

To show this step

$(k+1)^{th}$ term $\quad u_r = \frac{1}{4r^2-1}$

$$S_{k+1} = S_k + u_{k+1}$$

$$= S_k + \frac{1}{4(k+1)^2-1}$$

I.H

$$= f(k) + \frac{1}{4(k+1)^2-1}$$

$$= \frac{k}{2k+1} + \frac{1}{4(k+1)^2-1}$$

$$= \frac{4k(k+1)^2 - k + 2k + 1}{(2k+1)(4(k+1)^2-1)}$$

$$= \frac{4k^3 + 8k^2 + 4k + k + 1}{(2k+1)(4k^2+8k+3)}$$

$$= \frac{4k^3+8k^2+5k+1}{8k^3+20k^2+14k+3}$$

$$= \frac{(k+1)(4k^2+4k+1)}{(2k+3)(4k^2+4k+1)}$$ can force factorise

$$= \frac{k+1}{2k+3}$$

Competency skills

Q4. a) $S_1 = 1$ and $\frac{1}{2} \times 1 \times (1+1) = 1$

so base case true

b) $S_{k+1} = S_k + a_{k+1}$ series + new term

$= S_k + (k+1)$

$= \frac{1}{2}k(k+1) + (k+1) = (k+1)(\frac{1}{2}k+1)$

$= \frac{1}{2}(k+1)(k+2)$

induction step

so I.S is true

c) As the formula is true for $N=1$ and if true for k then it's true for $k+1$ (for any $k \geq 1$) we have by mathematical induction that the formula is true for all $N \geq 1$

Q5.

Base case: For $N=1$

$$S_1 = a_1 = 1^2 = 1$$

$$\frac{1}{6} \times 1 \times (1+1) \times (2 \times 1 + 1) = \frac{1}{6} \times 1 \times 2 \times 3 = 1$$

so base case holds

Induction hypothesis (I.H): Assume $S_k = \frac{1}{6}k(k+1)(2k+1)$ for some $k \geq 1$ then we have

$$S_{k+1} = \frac{1}{6}(k+1)(k+2)(2k+3)$$

Induction step (I.S):

$$S_{k+1} = S_k + a_{k+1}$$ series plus new term

I.H

$$= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2$$

$$= (k+1)\left(\frac{1}{6}k(2k+1) + (k+1)\right)$$

$$= \frac{(k+1)}{6}\left(2k^2 + k + 6k + 6\right)$$

$$= \frac{(k+1)}{6}\left(2k^2 + 7k + 6\right)$$

$$= \frac{(k+1)}{6}(2k+3)(k+2)$$

$$= \frac{1}{6}(k+1)(k+2)(2k+3).$$

Conclusion:

AS the formula is true for $N=1$ and if true for k then it's true for $k+1$ (for any $k \geq 1$) we have by mathematical induction that the formula is true for all $N \geq 1$

Q6. Base case: $N = 1$

$$\sum_{i=1}^{1} \frac{1}{i(i+1)} = \frac{1}{1 \times 2} = \frac{1}{2}$$

$\frac{1}{1+1} = \frac{1}{2}$ so Base case holds

I.H: Assume formula holds for some $k \geq 1$

series + term

$$\text{I.S:} \quad \sum_{i=1}^{k+1} \frac{1}{i(i+1)} = \sum_{i=1}^{k} \frac{1}{i(i+1)} + \frac{1}{(k+1)(k+2)}$$

I.H

$$= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{k(k+2)+1}{(k+1)(k+2)} = \frac{k^2+2k+1}{(k+1)(k+2)}$$

$$= \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

so formula holds for $k+1$.

Conclusion:

As the formula is true for $N=1$ and if true for k then it's true for k+1 (for any $k \geq 1$) we have by mathematical induction that the formula is true for all $N \geq 1$

Q7. $u_n = ar^{n-1}$ for $n \geq 1$

$$\sum_{n=1}^{N} u_n = \frac{a(1-r^N)}{1-r}$$

Base case : $N=1$

$$\sum_{n=1}^{1} u_n = a \qquad \frac{a(1-r^1)}{1-r} = a$$

so holds

I.H : Assume formula for some $k \geq 1$

I.S :

$$\sum_{n=1}^{k+1} u_n = \sum_{n=1}^{k} u_n + u_{k+1}$$

$$= \frac{a(1-r^k)}{1-r} + ar^k$$

$$= a\left(\frac{1-r^k}{1-r} + r^k\right)$$

$$= \frac{a}{1-r}\left(1-r^k + r^k - r^{k+1}\right)$$

$$= \frac{a(1-r^{k+1})}{1-r}$$ so holds for $k+1$

Conclusion:

As the formula is true for $N=1$ and if true for k then it's true for $k+1$ (for any $k \geq 1$) we have by mathematical induction that the formula is true for all $N \geq 1$

Advanced skills

Q8. Base case: $N=2$ ← Starts at 2

$$\left(1-\frac{1}{(r+1)^2}\right) = \left(1-\frac{1}{(1+1)^2}\right) = \frac{3}{4}$$

$$\frac{2+1}{2\times 2} = \frac{3}{4}$$

I.H: Assume formula holds for some

$k \geqslant 2$

Note. not a series but a product.

I.S: $\underbrace{\left(1-\frac{1}{2^2}\right)\cdots\left(1-\frac{1}{k^2}\right)}_{I.H}\left(1-\frac{1}{(k+1)^2}\right)$

$$= \frac{k+1}{2k} \times \left(1 - \frac{1}{(k+1)^2}\right)$$

$$= \frac{k+1}{2k} \times \frac{(k+1)^2-1}{(k+1)^2} = \frac{1}{2k} \times \frac{(k^2+1)-1}{k+1}$$

$$= \frac{(k+1)^2-1}{2k(k+1)} = \frac{k^2+2k}{2k(k+1)} = \frac{k+2}{2(k+1)}$$

So formula holds for k+2

Conclusion:

As the formula is true for N=1

and if true for k then it's true for

$k+1$ (for any $k \geq 1$) we have by mathematical induction that the formula is true for all $N \geq 1$

Q9. $1^3 + 2^3 + \ldots + n^3 = (1 + \ldots + n)^2$

Let $u_n = n^3$

Then statement/formula is

$$\sum_{n=1}^{N} n^3 = \left(\sum_{n=1}^{N} n\right)^2 = \left(\frac{1}{2}N(N+1)\right)^2$$

$$= \frac{1}{4}N^2(N+1)^2$$

Base case: $N = 1$

$$\sum_{n=1}^{1} n^3 = 1^3 = 1 \qquad \frac{1}{4} \times 1^2 \times 2^2 = 1$$

so holds

I.H: Assume formula holds for some $k \geq 1$

I.S: $$\sum_{n=1}^{k+1} n^3 = \sum_{n=1}^{k} n^3 + (k+1)^3$$

$$= \frac{1}{4}k^2(k+1)^2 + (k+1)^3$$

$$= \frac{1}{4}(k+1)^2\left(k^2 + 4(k+1)\right)$$

$$= \frac{1}{4}(k+1)^2(k+2)^2 \quad \text{so holds}$$

Conclusion:

As the formula is true for $N=1$ and if true for k then it's true for $k+1$ (for any $k \geq 1$) we have by mathematical induction that the formula is true for all $N \geq 1$

Q10. $1+3+5+\ldots+(2n-1)$

$N=1 \qquad 1=1$

$N=2 \qquad 1+3=4$

$N=3 \qquad 1+3+5=9$

$N=4 \qquad 1+3+5+7=15$

$\downarrow N^2$

so prove first N odd is N^2

Base case: Already seen

I.H: Assume first k odd no's sum to k^2 for some $k \geq 1$

I.S: Sum of first $k+1$ odd is

Sum of first k + $(k+1)^{th}$ odd no

$\downarrow$ I.H $\qquad\qquad$ $\downarrow$ term

$$k^2 + (2(k+1)-1)$$

$$k^2 + 2k + 1 = (k+1)^2$$

So holds

Conclusion:

As the statement is true for $N = 1$ and if true for k then it's true for $k+1$ (for any $k \geq 1$) we have by mathematical induction that the statement is true for all $N \geq 1$

Q11. Base case: $n=1$

$$1 = 1$$

holds

$$\frac{1-2x+x^2}{(1-x)^2} = 1$$

I.S: Assume holds for some $k \geq 1$

I.H:

$$1 + 2x + \dots + kx^{k-1} + (k+1)x^k$$

$\downarrow$ I.H

$$= \frac{1-(k+1)x^k + kx^{k+1}}{(1-x)^2} + (k+1)x^k$$

$$= \frac{1-(k+1)x^k + kx^{k+1} + (k+1)x^k(1-x)^2}{(1-x)^2}$$

$$= \frac{1-(k+1)x^k + kx^{k+1} + (k+1)x^k(1-2x+x^2)}{(1-x)^2}$$

$$= \frac{1 + (k - 2(k+1))x^{k+1} + (k+1)x^{k+2}}{(1-x)^2}$$

$$= \frac{1 - (k+2)x^{k+1} + (k+1)x^{k+2}}{(1-x)^2}$$

holds

Conclusion:

As the formula is true for $N = 1$ and if true for k then it's true for $k+1$ (for any $k \geq 1$) we have by mathematical induction that the formula is true for all $N \geq 1$